

SIMON MACKINNON

Senior Cloud Engineer | AWS | Terraform | Python | AI-Assisted Development

Yarraville, VIC, Australia | +61 (0) 447 063 082 | simon.mackinnon15@gmail.com

linkedin.com/in/simonmackinnon | theclouddevopslearningblog.com

SUMMARY

Senior Cloud & Infrastructure Engineer with 10+ years' experience delivering enterprise-scale AWS platforms across thousands of workloads. Experienced in AI-assisted development (Claude Code, Cursor, GitHub Copilot), cloud governance, IaC automation, and mentoring high-performance engineering teams.

SKILLS

Cloud & Infrastructure: AWS (multi-account, Landing Zones, Networking), Azure, Terraform, OpenTofu, CloudFormation, Harness IACM

Automation & DevOps: CI/CD (Jenkins, GitHub Actions), Configuration Management (Puppet), Infrastructure Pipelines

Programming: Python, Node.js, TypeScript, Bash, Lambda — automation tooling, governance scripts, compliance automation

AI Development: Claude Code, Cursor, GitHub Copilot, LLM API integration (Anthropic), AI pair programming

Architecture & Design: Cloud-Native Architecture, Cloud Governance, Cost Optimisation, Multi-account Design

Leadership: Mentoring, Stakeholder Engagement, Community Building, Change Leadership

EXPERIENCE

Senior Cloud Software Engineer | [National Australia Bank](#) | Oct 2025 – Present

- Contributing to an enterprise-scale migration from Terraform Enterprise to Harness IACM, and developing OpenTofu CDK modules as part of the organisation's shift to open-source IaC tooling.
- Leveraging AI-assisted development tooling (Cursor, Claude Code) to accelerate IaC delivery, and reverse-engineering existing AWS environments into version-controlled Terraform configuration.
- Integrated AI coding workflows into team practices, publishing write-ups on real-world productivity gains and limitations of AI pair programming.

Senior Cloud Engineer – AWS SME | [National Australia Bank](#) | Jul 2023 – Sep 2025

- Designed and deployed scalable, multi-account AWS IaC frameworks, enabling rapid provisioning for 100+ business units and thousands of workloads with improved reliability and security posture.
- Automated key operational workflows in Python and Jenkins, reducing manual maintenance effort by over 50%.
- Championed cloud governance and architectural best practices, enabling secure, compliant, and auditable infrastructure growth across the organisation.

Key Achievements:

- Engineered a bulk Terraform Plan & Apply platform (Terraform, Jenkins EC2 plugin, Bash) enabling consistent infrastructure changes across 100+ AWS accounts in under one hour — previously a multi-day process.
- Designed and delivered NAB's AWS RDS Compliance Manager — a cloud-native, event-driven governance platform providing centralised compliance controls across 3,000+ RDS instances.

Cloud Engineer | [National Australia Bank](#) | Feb 2019 – Jul 2023

- Delivered core AWS landing zone architecture, accelerating provisioning time by 40% and improving compliance posture.
- Built automated compliance and reporting pipelines, cutting regulatory reporting time from weeks to hours.
- Led the AWS Community of Practice and tutored engineers through the LearnToCode@NAB programme, mentoring dozens in modern software and cloud fundamentals.

Software Developer | [MLC Life Insurance](#) | Jan 2016 – Jan 2019

- Full-stack development (jQuery, .NET, SQL Server), CI/CD pipelines (Jenkins), config-as-code (Puppet). Co-founded a developer community of practice.
- Administered Windows Server environments — IIS web server configuration and deployment, and Puppet-managed configuration of Windows infrastructure.

Earlier Roles

Software Engineer – Redflex Traffic Systems (Nov 2013 – Oct 2015) – Radar and traffic camera systems in C++ and .NET.

Software & Electronics Engineer – Ingeneus Pty Ltd (Jan 2012 – Oct 2013) – Medical device firmware and iOS apps in Python, C, C++ and Objective-C.

Software Developer Intern – Flanders’ Mechatronics Centre, Belgium (Jul–Nov 2009) – COTS and open-source industrial control software integrations.

SIDE PROJECTS

TypeStar – Free kids’ typing tutor. React 18 / TypeScript / Vite, AWS (S3, CloudFront, Cognito, API Gateway, Lambda, DynamoDB), Terraform, ElevenLabs audio, GitHub Actions CI/CD. Built with Claude Code.

typestar.theclouddevopslearningblog.com

Kids Code Realm – Medieval fantasy-themed coding education site for children.

coderealm.theclouddevopslearningblog.com

MemeCloud – Serverless meme generator. Lambda (Python/Pillow), S3, CloudFront, API Gateway, DynamoDB. Built with GitHub Copilot then Claude Code; wrote a candid comparison of both. memes.theclouddevopslearningblog.com

BreweriesNearMe – Static SPA on S3 + CloudFront + Route 53 with a Python data scraper for the OpenBreweryDB dataset. breweriesnearme.theclouddevopslearningblog.com

OPEN SOURCE

Open Brewery DB – github.com/openbrewerydb/openbrewerydb

- Extended the API to support radius-based filtering on the by_dist endpoint, enabling location-scoped distance queries.
- Built data scrapers and contributed brewery datasets for Australia, New Zealand, Germany, and Belgium, expanding international database coverage.

EDUCATION

Bachelor of Engineering (Hons) – Robotics & Mechatronics | Bachelor of Science – Computer Science & Software Engineering

Swinburne University of Technology | 2006 – 2011

CERTIFICATIONS

Current: Terraform Associate – HashiCorp (2024) | **In Progress:** AWS DevOps Engineer – Professional (2026), AWS CloudOps Engineer – Associate (renewal 2026)

Previous: AWS SysOps Administrator – Associate, AWS Developer Associate, AWS Solution Architect Associate (all 2019), Azure Fundamentals (2021)

VOLUNTEERING & COMMUNITY

Volunteer Tutor, Kensington Neighbourhood House (2015–16), tutoring 30+ secondary students in computing. | Founded and ran { Dev : CoP }, a lean-coffee tech meetup for up to 60 IT professionals. | Co-organised the Melbourne AWS Programming & Tools Meetup, with in-person events drawing up to 200 cloud practitioners.

References available on request